AICTE – EDUSKILLS VIRTUAL INTERNSHIP

# Google AI-ML

An 8-Week Virtual Internship Presentation Report

*Internship Duration: June 2026 – August 2026 (8 Weeks)*


**ANANT RAJ**

Roll No: 2410031035 | Batch: 2024-28

B.Tech (CSE) • **IILM University, Greater Noida, U.P.**


Supported by

**Google for Developers**

# Candidate & Program Overview

## Candidate Details

Student Name

**ANANT RAJ**

Roll Number

**2410031035**

Degree

**B.Tech (Computer Science and Engineering)**

Academic Batch

**2024-28**

University

**IILM University, Greater Noida, U.P.**

## Internship Profile

Program Name

**AICTE – EduSkills Virtual Internship**

Specialization Domain

**Google AI-ML**

Duration

**8 Weeks (June 2026 – August 2026)**

AICTE Student ID

**STU69986302ad9bd1771594498**

Offer Letter Ref No.

**C17Y2026M061447012**

# Organization Profile: EduSkills Academy

EduSkills Academy is the ISO 9001:2015 / ISO 20000-1:2018 certified body that ran this internship, under the tagline “Nation Building Through Skills.” It works with AICTE to deliver the AICTE National Internship Portal.

### Google for Developers Partnership

For the Google AI-ML track specifically, EduSkills doesn't write its own material – it plugs straight into Google's public learning pathways and codelabs under the India Edu Program.

### Credentialing & Verification

Pass the quiz for a pathway and you get a real, verifiable Google Developer badge on a public profile, not just an internal EduSkills record.

### AICTE National Internship Portal

Runs under the Ministry of Education, Government of India, so the internship carries formal academic recognition for engineering students across the country.

### Focus Domains

Beyond AI/ML, EduSkills also runs tracks in Cloud Computing, Cybersecurity, and Applied Data Science, all built the same way – on top of official industry pathways.

# Problem Statement & Industry Context

## The Industry Gap

Most of us learn the maths behind Machine Learning in class – gradients, loss functions, backprop – but rarely connect it to a working app. Ideas like neural networks, object detection, and image classification tend to stay abstract, with no clear route from “trained model” to “something on a phone.”

## The Internship's Answer

This program pushed past that stopping point. It took a model from TensorFlow training, through TensorFlow Lite optimization, into on-device inference via ML Kit, and finally into a Cloud Vision API-backed product search – a real, mobile-ready pipeline, start to finish.

# Core Project Objectives

### Neural Networks with TensorFlow

Go from ML first principles to CNNs used for image recognition.

### Object Detection

Start with pretrained detectors, then train custom ones with TensorFlow Lite.

### Product Image Search

Build a mobile search feature powered by on-device object detection.

### Image Classification

Build, test, and keep improving custom image classifiers.

### Developer Credentialing

Turn each completed pathway into a real, verifiable Google badge.

### Public Developer Profile

Publish a g.dev/ profile so the badges are actually visible to others.

# Technologies & ML Toolchain

### ML Framework

- TensorFlow
- TensorFlow Lite
- Model Maker Library

### Mobile Integration

- Google ML Kit
- Object Detection API
- Android Application Layer

### Cloud & Vision

- Google Cloud Vision API
- Product Search Backend
- Google Colab / Codelabs

### Credentialing

- Google Developer Profile
- EduSkills Assess2ME
- g.dev/ Public Web Address

# On-Device AI/ML Architecture

**Model Layer** — TensorFlow neural network trained from first principles to CNNs

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**Optimization Layer** — TensorFlow Lite + Model Maker compress the model for mobile/edge use

↓

**Inference Layer** — Google ML Kit runs the model on-device inside the Android app

↓

**Backend Layer** — Cloud Vision API – Product Search matches images to a product catalogue

↓

**Application Layer** — Mobile app calls the model / backend and returns results to the user

↓

**Credentialing Layer** — Passed quizzes issue Google Developer badges to a public profile

# Curriculum Timeline (Weeks 1 to 4)

**Week 1**

### Program Neural Networks with TensorFlow

ML first principles to CNNs for image recognition.

**Week 2**

### Get Started with Object Detection

Integrate pretrained object detectors into mobile apps.

**Week 3**

### Go Further with Object Detection

Train custom detectors with TensorFlow Lite & Model Maker.

**Week 4**

### Get Started with Product Image Search

Build product image search using on-device detection.

# Curriculum Timeline (Weeks 5 to 8)

**Week 5**

### Get Started with Product Image Search

Detect objects in static images & live camera feeds.

**Week 6**

### Go Further with Product Image Search

Build backend integration for the mobile application.

**Week 7**

### Go Further with Image Classification

Build custom classifiers, improve classification skill.

**Week 8**

### Go Further with Image Classification

Integrate a custom image classifier into the app.

# Key Projects & Demonstrated Skills

- Started with the “Hello World of Machine Learning” in TensorFlow and worked up to convolutional neural networks for image recognition.
- Got a pretrained object detector running inside a mobile app using Google ML Kit's Object Detection API, live and on-device.
- Trained my own custom object-detection model from scratch using TensorFlow Lite and the Model Maker library.
- Built a product image search feature that picks out objects in both still photos and a live camera feed.
- Hooked that search feature up to a real backend, built on the Google Cloud Vision API – Product Search.
- Iterated on custom image-classification models until I had one actually working inside a mobile app.
- Published a public Google Developer Profile at a custom g.dev/ address, with every badge earned on display.

# Certification & Official Verification

| | |
|---|---|
| **Internship Title** | Google AI-ML Virtual Internship |
| **Issuing Organization** | EduSkills Academy (ISO 9001:2015 & ISO 20000-1:2018 Certified) |
| **Academic Patronage** | AICTE National Internship Portal & EduSkills Foundation |
| **Certificate ID** | 47ffa0784e65a7fcc6e1 |
| **Offer Letter Ref No.** | C17Y2026M061447012 |
| **AICTE Student ID** | STU69986302ad9bd1771594498 |
| **Final Grade** | “O” — Outstanding (90–100) |

# Internship Completion Certificate

शिक्षा मंत्रालय
MINISTRY OF EDUCATION

NATIONAL INTERNSHIP PORTAL

EduSkills
Nation Building Through Skills

AICTE - EduSkills 2026 VIRTUAL INTERNSHIP

## Certificate of Virtual Internship

This is to certify that

Anant Raj

IILM University, Greater Noida

has successfully completed the 8-weeks

**AI-ML Virtual Internship**

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By : India Edu Program
Google for Developers

Karthik Padmanabhan
Developer Ecosystem Lead
MENA & India, Google

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 47ffa0784e65a7fcc6e1
Student ID: STU69986302ad9bd1771594498

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

# Thank You!

**ANANT RAJ**

Roll No: 2410031035

**School of Computer Science & Engineering**

IILM University, Greater Noida, U.P.